

while knowing the forces being applied by the end effector.

Sometimes more than one type of sensory input is used to increase the feel factor. We call it multimodal interaction method. Haptic information can be used as the primary source of information to control your system, or it can be used as a secondary information channel. Secondary signals may be like warning signals that a potential dangerous situation is being approached. The warning tone on your PC when you give some wrong command with your mouse is a prime example of secondary haptic audio information being provided to your ears.

What are the key components of a haptic-based system?

The key components would be actuators, sensors, effectors, controllers and drive circuits, software and displays for the user interface. Different types of actuation mechanisms can be used to simulate different types of sensory inputs to the human body. For example, electrostatic technology can be used to simulate friction and normal indentation. Piezoelectric technology can be used to simulate normal indentation, lateral skin stretch and vibration.

Similarly, a particular feel can be simulated by different technologies. For example, normal indentation can be simulated by electrostatic and piezoelectric motors, shape memory alloys, electromagnetic micro coils, air jets, pneumatic valves and dimples, acoustic radiation pressure and electro simulation.

What are the main design parameters for haptic effect?

The main design parameters are duration (how long the haptic effect will last), envelope (the area that bounds the attack (start) and fade (end) portions of an effect and allows you to change the level of the effect over its duration), magnitude (strength of the output of the sensory perception actuator), period (timing between sensation pulses), phase (point at which the effect begins with respect to an event) and waveform (shape of the repeated sensation pulses).

What are the key design challenges in designing a haptic-based system?

The main challenge is to make a control interface that feels exactly like the tool originally being used by the operator. Take the case of a haptic device that mimics the tools of an endoscopic surgeon. The haptic device has to feel like a normal endoscope. The actuators and sensors have to be implemented to look almost like that. The major issue would be balancing the weight of the different actuators, etc along the different axes so that the surgeon feels like he is using his original equipment. This is called muscle memory. The surgeon does not have to re-train himself to work on this machine.

Another design aspect would be to provide the kinetic motion and freedom exactly like the surgeon would feel while conducting the said operation on an actual patient. It is also very important to make the software and control mechanism fast and accurate.

Could you please explain with some example?

When we were designing a haptic device for surgical simulation, one of the major challenges was balancing the weight of all the mechanical components and nullifying the imbalances of the weight. When an actuator is used, it adds weight to the device. So we had to design the structure in such a way that the weights cancel out.

Secondly, we wanted to give a natural motion path to the operation of the device. If endoscopic operation is considered, we have to actually follow a path that a doctor in actual situation would take. So the whole device has to be made such that the doctor feels like he is actually operating.

We had to juggle the mechanical designs and the software feedback to actively overcome the weight with the software. We came up with a device that had five degrees of motion, of which only one degree of motion had to be actively stabilised whereas the remaining four degrees were naturally stabilised as far as the weight and friction were concerned.

What are the new applications emerging for haptic-based systems?

Haptics can be used anywhere depending on your imagination. When there is an operator involved in commanding a robotic system where application of forces is critical, a haptic device becomes a must. In a normal system, joysticks are used and the only thing known is the extent of the motion, not the amount of force.

We are using haptics for mobile robotic system for bomb disposal and surgical simulation, where the doctor can 'practise' a surgery that he is supposed to do on a live patient, or try to simulate different ways in which he can perform the same surgery from different locations. We are trying to use haptic devices for assembly simulation. Most of the haptic devices that we have developed are passive devices or active devices where, like motor actuators, the device pushes an operation being performed.

Take the case of vehicular route mapping devices. Presently, the driver has to look at the device to know where to turn. The device can be made to have a directional vibrating mode where direction and amplitude of the vibration are dependent on the nearness of the turn. The driver can place the device in his pocket. As a turn approaches, the directional vibration will tell the driver about the turn coming up and the direction. He does not have to take his eyes off the road, which will be much safer for him and others on the road.

Another main application area is controlling an exoskeleton strapped to the human body. Especially useful for a soldier, this will make it easier for him to carry larger loads and control heavier machines. Haptics is also being used to mimic terrain realism.

Haptics is being used extensively to teach surgery and simulate different surgical procedures. ●

(The complete interview can be viewed at electronicsforu.com)